



Homotopy Analysis of Unsteady Magnetohydrodynamic Two-Phase Casson Nanofluid Flow Over a Bidirectional Stretching Surface With Cattaneo–Christov Heat Flux, Thermal Radiation, and Double-Diffusive Effects

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Abstract: The study of non-Newtonian nanofluid flow over magnetohydrodynamic (MHD) surfaces has attracted considerable research interest due to its relevance to polymer processing, heat treatment, metallurgical production, and biomedical transport systems. Despite extensive studies on MHD Casson nanofluids, the combined effects of bidirectional stretching, porous-medium resistance, thermal radiation, viscous dissipation, Brownian motion, thermophoresis, double diffusion, and non-Fourier heat conduction have received relatively limited attention within a unified analytical framework. To address this gap, this study develops an analytical model for the unsteady flow of MHD Casson nanofluids over a bidirectional stretching surface by incorporating the Cattaneo–Christov heat-flux formulation. The proposed model accounts for finite-speed thermal propagation and thermal relaxation effects that are not considered in the conventional Fourier heat conduction model. The resulting nonlinear partial differential equations are transformed into a coupled system of similarity equations and analysed using the Homotopy Analysis Method (HAM). The effects of the governing parameters on the velocity, temperature, and nanoparticle concentration profiles are systematically analysed. The results show that stronger magnetic forces and increased permeability of the porous medium reduce the fluid velocity, whereas thermal radiation and viscous dissipation enhance the temperature distribution. Brownian motion and thermophoresis influence nanoparticle concentration and thermal boundary-layer characteristics, while increasing the thermal relaxation parameter in the Cattaneo–Christov model reduces the temperature profile compared with the classical Fourier formulation, highlighting the effects of finite-speed heat propagation and thermal relaxation. The analytical results provide comprehensive insights into the coupled momentum, heat, and mass transfer characteristics of cryogenic nanofluids and may contribute to the design and optimisation of thermal management systems and polymer-processing applications.

Keywords: Magnetohydrodynamic; Casson fluid; Cattaneo–Christov heat flux; Bidirectional stretching surface; Homotopy Analysis method; Heat and mass transfer

1 Introduction

The increasing demand for effective thermal management in modern engineering applications has led to extensive research into advanced heat- and mass-transfer fluids. Conventional heat-transfer fluids, such as water, ethylene glycol, and mineral oils, typically exhibit relatively low thermal conductivity, which limits their effectiveness in systems subjected to high heat fluxes [1]. Nanofluids have therefore attracted considerable attention as alternative working fluids because the incorporation of nanoparticles can improve thermal conductivity and enhance overall heat-transfer performance [2]. As a result, nanofluids have been increasingly employed in areas such as electronic cooling, renewable-energy systems, biomedical engineering, chemical processing, and advanced manufacturing [3, 4].

Magnetohydrodynamic (MHD) flow has also received substantial research attention because an externally imposed magnetic field provides an effective means of controlling the motion of electrically conducting fluids [5].

The interaction between the applied magnetic field and the electric currents within the conducting fluid produces a Lorentz force that alters the hydrodynamic and thermal characteristics of the flow. This mechanism of flow regulation is relevant to several applications, including metallurgical processing, crystal growth, electromagnetic casting, nuclear-reactor cooling, geothermal-energy extraction, and biomedical transport. Recent studies have further highlighted the importance of magnetic-field effects in modifying the thermal and flow characteristics of complex non-Newtonian and nanofluid systems under different operating conditions [6, 7]. Among the various rheological models used to describe non-Newtonian fluids, the Casson model is particularly significant because it accounts for yield-stress characteristics, whereby a critical stress must be exceeded before fluid motion occurs [8]. The model has been applied successfully to physiological fluids such as blood and to industrial materials including printing inks, polymer melts, paints, chocolate, and cosmetic products [9]. The combined influence of Casson rheology and magnetic fields is consequently of considerable interest in biomedical applications, polymer processing, and advanced-material manufacturing [10]. Incorporating magnetic effects into the Casson-fluid formulation therefore provides an appropriate framework for analysing electrically conducting non-Newtonian flows encountered in these applications [11].

Previous investigations have shown that the interaction between yield-stress behaviour and magnetic forces can substantially influence the hydrodynamic characteristics of Casson fluids, particularly their velocity distributions and wall shear stresses [12]. The inclusion of porous-medium resistance further intensifies these interactions, leading to notable changes in fluid motion and thermal transport characteristics [13]. Such effects are particularly relevant to engineering systems in which fluid movement occurs through permeable structures or porous materials.

Boundary-layer flow over stretching surfaces has long been an important area of research because of its relevance to numerous industrial processes, including polymer extrusion, metal spinning, continuous casting, glass-fibre production, wire drawing, and coating operations [14]. Early studies primarily examined steady flow generated by one-dimensional stretching surfaces and established the fundamental principles governing momentum and heat transfer in these configurations [15]. Subsequent research incorporated unsteady stretching to account for temporal variations in surface motion and their influence on the associated transport processes [16]. More recent investigations have extended these formulations to bidirectionally stretching surfaces, providing a more representative description of manufacturing processes involving simultaneous stretching along two spatial directions [17]. These developments have contributed significantly to the understanding of momentum, heat, and mass transport in fluid systems subjected to multidirectional and time-dependent stretching conditions [18].

An accurate description of heat transfer also requires an appropriate constitutive model for thermal transport, particularly when finite-speed propagation and thermal relaxation effects become important. The conventional Fourier law assumes instantaneous propagation of thermal disturbances, an assumption that may be inadequate for high-speed flows, microscale thermal systems, and materials exhibiting significant thermal relaxation. The Cattaneo–Christov heat-flux model overcomes this limitation by incorporating a thermal relaxation time into the energy equation, thereby permitting heat propagation at a finite speed [19]. This non-Fourier formulation has subsequently been applied to various non-Newtonian flow problems in which thermal relaxation effects are important [20]. Its use has also been extended to nanofluid systems, where the interaction between nanoparticle transport and thermal relaxation can significantly influence the resulting heat-transfer characteristics [21].

For nonlinear boundary-layer problems involving strongly coupled momentum, energy, and concentration equations, analytical and semi-analytical solution techniques are particularly useful. The homotopy analysis method (HAM) provides an effective analytical framework for such problems because it generates convergent series solutions without requiring the presence of a small or large physical parameter. Zhao et al. [22] demonstrated the applicability of HAM to three-dimensional nanofluid flow and heat transfer over a surface undergoing simultaneous stretching in two directions, obtaining analytical representations of the corresponding velocity and temperature fields. Their work, published in *Boundary Value Problems*, provides an important foundation for the application of HAM to multidirectional nanofluid flow problems.

In addition to the aforementioned mechanisms, several other physical effects can significantly influence heat and mass transport in engineering flow systems. Thermal radiation becomes increasingly important in high-temperature applications, including solar-energy systems, combustion chambers, gas turbines, and metallurgical furnaces [23]. Viscous dissipation accounts for the conversion of mechanical energy into internal thermal energy and can therefore substantially modify the temperature field, particularly in flows characterized by strong velocity gradients [24]. Within nanofluid systems, Brownian motion contributes to nanoparticle dispersion and influences the associated thermal transport processes [25]. Thermophoresis, meanwhile, drives nanoparticles from regions of higher temperature toward regions of lower temperature, thereby modifying nanoparticle concentration and the thickness of the concentration boundary layer [26]. The combined effects of these mechanisms are particularly important in the analysis of nanofluid transport through porous structures [27] and may become more pronounced when the nanofluid is electrically conducting and exposed to an external magnetic field [28]. Consequently, incorporating these coupled physical effects into a unified mathematical framework is essential for obtaining a more comprehensive

understanding of the momentum, heat, and mass-transfer behaviour of MHD Casson nanofluids.

Despite considerable progress in the investigation of MHD Casson nanofluid flows, most existing studies have focused on specific combinations of the relevant physical mechanisms rather than their comprehensive integration. Only a limited number of investigations have simultaneously considered unsteadiness, bidirectional stretching, porous-medium resistance, thermal radiation, viscous dissipation, Brownian motion, thermophoresis, double-diffusive transport, and the Cattaneo–Christov heat-flux model within a single analytical framework. Moreover, analytical studies employing the HAM for such an extensive combination of physical effects remain relatively scarce.

In response to this research gap, the present study investigates unsteady MHD Casson nanofluid flow over a bidirectionally stretching surface while incorporating porous-medium resistance, thermal radiation, viscous dissipation, Brownian motion, thermophoresis, double-diffusive transport, and Cattaneo–Christov heat conduction. The governing nonlinear partial differential equations are reduced to a coupled system of similarity equations and subsequently solved analytically using the HAM. In contrast to the conventional Fourier heat-conduction model, the Cattaneo–Christov formulation incorporates thermal relaxation and finite-speed heat propagation, thereby providing a more physically representative description of heat transfer in media with significant relaxation effects. The resulting framework offers deeper insight into the coupled momentum, heat, and mass-transfer processes occurring in electrically conducting Casson nanofluids and may provide useful guidance for the development and optimisation of thermal-management systems, biomedical applications, polymer-processing technologies, and advanced manufacturing processes.

A two-phase nanofluid formulation is adopted in the present investigation to provide a more realistic description of nanoparticle transport than the conventional single-phase approach, which generally assumes thermal equilibrium and identical velocities between the nanoparticles and the base fluid. Within the adopted formulation, nanoparticle migration is primarily governed by Brownian diffusion and thermophoresis, represented by the Brownian motion (Nb) and thermophoresis (Nt) parameters, respectively. Incorporating these mechanisms enables a more detailed representation of the coupled heat- and mass-transfer processes and improves the prediction of temperature and concentration distributions, particularly in flows characterized by strong thermal gradients and moderate nanoparticle volume fractions where relative particle-fluid motion may become appreciable. Nevertheless, the model assumes a dilute, uniformly dispersed nanoparticle suspension and does not account for particle agglomeration, sedimentation, or other inter-particle interactions. Consequently, the adopted two-phase formulation provides a reasonable compromise between physical representation and mathematical tractability for the analysis of the present unsteady MHD Casson nanofluid flow with Cattaneo–Christov heat flux. The nanoparticle transport framework employed in this study is based on the formulation proposed by Buongiorno [29], which identifies Brownian diffusion and thermophoresis as the dominant mechanisms responsible for nanoparticle slip relative to the base fluid. This framework has subsequently served as an important theoretical basis for investigating coupled nanoparticle heat and mass transport across a broad range of nanofluid flow configurations.

2 Problem Origination

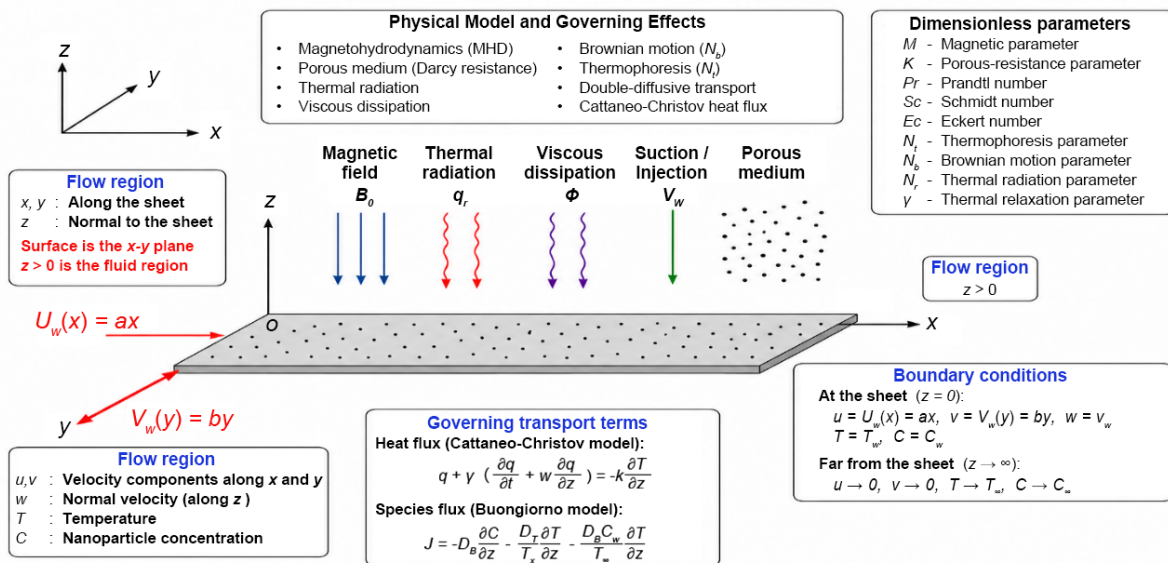


Figure 1. Flow geometry

We cogitate an unsteady, three-dimensional MHD flow of a two-phase Casson nanofluid over a bidirectionally stretching surface. The flow is assumed to be incompressible, laminar, and electrically conducting. A uniform magnetic field of strength B_0 is applied normal to the stretching surface. The induced magnetic field is neglected under the assumption of a low magnetic Reynolds number. The physical configuration of the flow is illustrated in Figure 1. The stretching surface is assumed to move in two mutually perpendicular directions with velocity components proportional to the spatial coordinates, such that: $u_w = ax$, $v_w = by$, where a and b are the stretching rates along the x - and y -directions, respectively. The fluid occupies the region $z > 0$, and the flow is generated due to the stretching of the surface. The nanofluid is modeled as a two-phase system consisting of a base fluid and dispersed nanoparticles, with nanoparticle transport governed by Brownian diffusion and thermophoresis. The rheological behaviour of the fluid is described using the Casson fluid model, which accounts for yield stress effects. Furthermore, the heat transfer mechanism is governed by the Cattaneo–Christov heat flux model, which introduces thermal relaxation time to overcome the limitations of the classical Fourier law.

Under the above assumptions, the governing equations for conservation of mass, momentum, energy, and nanoparticle concentration can be written as follows.

3 Methodology

Continuity Equation

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \quad (1)$$

Momentum Equations

x -direction (Primary Velocity)

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} = \frac{\mu}{\rho} \left(1 + \frac{1}{\beta} \right) \frac{\partial^2 u}{\partial z^2} - \left(\frac{\sigma_T B_0^2}{\rho} + \frac{\mu}{\rho K_p} \right) u \quad (2)$$

y -direction (Secondary Velocity)

$$\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} = \frac{\mu}{\rho} \left(1 + \frac{1}{\beta} \right) \frac{\partial^2 v}{\partial z^2} - \left(\frac{\sigma B_0^2}{\rho} + \frac{\mu}{\rho K_p} \right) v \quad (3)$$

where, β is the Casson parameter, σ is electrical conductivity, ρ is fluid density, and K_p is the permeability parameter of the porous medium. The corresponding dimensionless porous-resistance parameter is defined by $K = v_f / a K_p$.

Energy Equation (Cattaneo–Christov Model)

$$\left. \begin{aligned} \frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + w \frac{\partial T}{\partial z} &= \frac{k}{(\rho C_p)} \frac{\partial^2 T}{\partial z^2} + \frac{16\sigma^* T_\infty^3}{3k^*(\rho C_p)} \frac{\partial^2 T}{\partial z^2} - \lambda \left(\frac{DT}{Dt} \right) + \\ &\frac{q_1}{(\rho C_p)_T} (T - T_\infty) + \frac{q_2}{(\rho C_p)} (T_f - T_\infty) \ell^{-an} + \frac{\sigma B_0^2}{(\rho C_p)} u^2 + \frac{\sigma}{(\rho C_p)} \left(\frac{\partial u}{\partial z} \right)^2 \end{aligned} \right\} \quad (4)$$

where, λ is the thermal relaxation time, and the last terms represent viscous dissipation (Eckert effect) and heat source/sink.

Nanoparticle Concentration Equation

$$\frac{\partial C}{\partial t} + u \frac{\partial C}{\partial x} + v \frac{\partial C}{\partial y} + w \frac{\partial C}{\partial z} = D_B \frac{\partial^2 C}{\partial z^2} + \frac{D_T}{T_\infty} \frac{\partial^2 T}{\partial z^2} \quad (5)$$

where, D_B and D_T denote Brownian motion and thermophoretic diffusion coefficients, respectively.

Boundary conditions are:

At the stretching surface ($z = 0$):

$$\left. \begin{aligned} u &= u_w + A^* u_z, v = v_w + A^* v_z, w = 0 \\ -k T_z &= h_f (T_f - T) \\ C &= C_w \end{aligned} \right\} \quad (6)$$

At $z \rightarrow \infty$:

$$u \rightarrow 0, v \rightarrow 0, T \rightarrow T_\infty, C \rightarrow C_\infty \quad (7)$$

The set of appropriate variables are:

$$u_w = \frac{ax}{1 - ct}, v_w = \frac{by}{1 - ct} \quad (8)$$

where, a, b : stretching rates; c : stretching rates.

$$\left. \begin{aligned} \eta &= z\sqrt{\frac{a}{v_f(1-ct)}}, u = \frac{ax}{1-ct}f'(\eta), v = \frac{by}{1-ct}g'(\eta), w = -\sqrt{\frac{av_f}{(1-ct)}}[f(\eta) + g(\eta)] \\ \theta(\eta) &= \frac{T-T_\infty}{T_f-T_\infty}, \phi(\eta) = \frac{C-C_\infty}{C_w-C_\infty} \end{aligned} \right\} \quad (9)$$

Incorporating Eq. (9) into Eqs. (1) – (7), we get:

$$\left(1 + \frac{1}{\beta}\right) f''' + (f + g)f'' - (f')^2 - S\left(f' + \frac{\eta}{2}f''\right) - (M + K)f' = 0 \quad (10)$$

$$\left(1 + \frac{1}{\beta}\right) g''' + (f + g)g'' - (g')^2 - S\left(g' + \frac{\eta}{2}g''\right) - (M + K)g' = 0 \quad (11)$$

$$\begin{aligned} (1 + R)\theta'' + Pr(f + g)\theta' - PrS\left(\theta + \frac{\eta}{2}\theta'\right) + PrEc(f''^2 + g''^2) \\ + PrMEc(f'^2 + g'^2) + Nb\theta'\phi' + Nt(\theta')^2 - \gamma[(f + g)\theta'' + f'\theta'] = 0 \end{aligned} \quad (12)$$

$$\phi'' + Sc(f + g)\phi' - ScS\left(\phi + \frac{\eta}{2}\phi'\right) + \frac{Nt}{Nb}\theta'' = 0 \quad (13)$$

The transformed Boundary conditions:

At $\eta = 0$;

$$\left. \begin{aligned} f'(0) &= 1 + \delta_1 f''(0), \quad g'(0) = \lambda + \delta_2 g''(0), \quad f(0) + g(0) = 0 \\ -\theta'(0) &= Bi(1 - \theta(0)) \\ \phi(0) &= 1 \end{aligned} \right\} \quad (14)$$

At $\eta \rightarrow \infty$;

$$f' \rightarrow 0, g' \rightarrow 0, \theta \rightarrow 0, \phi \rightarrow 0 \quad (15)$$

where, M denotes the magnetic parameter and K denotes the dimensionless porous-resistance parameter, here, Kp is the dimensional permeability of the porous medium. Thus, larger K represents stronger porous resistance, whereas larger Kp represents greater permeability and consequently weaker porous resistance. S stands for Unsteadiness parameter, Pr stands for Prandtl number, Rd stands for Radiation parameter, Ec stands for Eckert number (viscous dissipation), Nb stands for Brownian motion, Nt stands for Thermophoresis, λ stands for Thermal relaxation (Cattaneo–Christov).

3.1 Continuity Equation

To apply HAM to Eqs. (10)–(13) with boundary conditions Eqs. (14)–(15), we first define suitable initial guesses:

$$f_0(\eta) = 1 - \ell^{-\eta}, g_0(\eta) = 1 - \ell^{-\eta}, \theta_0(\eta) = \ell^{-\eta}, \phi_0(\eta) = \ell^{-\eta} \quad (16)$$

These satisfy the conditions:

$$\left. \begin{aligned} f'(0) &= 1, g'(0) = 1, \theta(0) = 1, \phi(0) = 1, \\ f'(\infty) &= 0, g'(\infty) = 0, \theta(\infty) = 0, \phi(\infty) = 0 \end{aligned} \right\} \quad (17)$$

We define linear operators:

$$L_f[f] = f''' - f', L_g[g] = g''' - g', L_\theta[\theta] = \theta'' - \theta', L_\phi[\phi] = \phi'' - \phi \quad (18)$$

Zeroth-Order Deformation Equations

Let $p \in [0, 1]$ be the embedding parameter and $\eta_f, \eta_g, \eta_\theta, \eta_\phi$ be convergence-control parameters.

We define:

$$\left\{ \begin{aligned} (1 - p)L_f[f(\eta, p) - f_0(\eta)] &= p\eta_f \mathcal{N}_f[f, g] \\ (1 - p)L_g[g(\eta, p) - g_0(\eta)] &= p\eta_g \mathcal{N}_g[f, g] \\ (1 - p)L_\theta[\theta(\eta, p) - \theta_0(\eta)] &= p\eta_\theta \mathcal{N}_\theta[\theta, f, g, \varphi] \\ (1 - p)L_\phi[\varphi(\eta, p) - \varphi_0(\eta)] &= p\eta_\phi \mathcal{N}_\phi[\varphi, \theta] \end{aligned} \right. \quad (19)$$

Boundary Conditions for Deformation:

$$\left. \begin{aligned} f'(0; p) = 1, g'(0; p) = 1, \theta(0; p) = 1, \phi(0; p) = 1 \\ f'(\infty; p) = 0, g'(\infty; p) = 0, \theta(\infty; p) = 0, \phi(\infty; p) = 0 \end{aligned} \right\} \quad (20)$$

Series Expansion

Assume solutions in power series of p :

$$\left. \begin{aligned} f(\eta; p) &= f_0(\eta) + \sum_{m=1}^{\infty} f_m(\eta) p^m \\ g(\eta; p) &= g_0(\eta) + \sum_{m=1}^{\infty} g_m(\eta) p^m \\ \theta(\eta; p) &= \theta_0(\eta) + \sum_{m=1}^{\infty} \theta_m(\eta) p^m \\ \phi(\eta; p) &= \phi_0(\eta) + \sum_{m=1}^{\infty} \phi_m(\eta) p^m \end{aligned} \right\} \quad (21)$$

At $p = 1$:

$$\left. \begin{aligned} f(\eta) &= f_0 + \sum_{m=1}^{\infty} f_m \\ g(\eta) &= g_0 + \sum_{m=1}^{\infty} g_m \\ \theta(\eta) &= \theta_0 + \sum_{m=1}^{\infty} \theta_m \\ \phi(\eta) &= \phi_0 + \sum_{m=1}^{\infty} \phi_m \end{aligned} \right\} \quad (22)$$

Differentiating with respect to p and setting $p = 0$:

$$\left\{ \begin{aligned} L_f [f_m - \chi_m f_{m-1}] &= \eta_f R_m^f(\eta) \\ L_g [g_m - \chi_m g_{m-1}] &= \eta_g R_m^g(\eta) \\ L_\theta [\theta_m - \chi_m \theta_{m-1}] &= \eta_\theta R_m^\theta(\eta) \\ L_\phi [\phi_m - \chi_m \phi_{m-1}] &= \eta_\phi R_m^\phi(\eta) \end{aligned} \right. \quad (23)$$

where,

$$\chi_m = \begin{cases} 0, & m \leq 1 \\ 1, & m > 1 \end{cases} \quad (24)$$

And the residual functions $R_m^f, R_m^g, R_m^\theta, R_m^\phi$ come from the nonlinear terms of Eqs. (10)–(13) respectively. By using the standard HAM auxiliary linear operators, from Eq. (23), the operators are of the form:

$$L_f[f] = f''' - f', L_g[g] = g''' - g', L_\theta[\theta] = \theta'' - \theta', L_\phi[\phi] = \phi'' - \phi \quad (25)$$

So their inverse naturally produces exponential solutions:

$$f_0(\eta) = 1 - \ell^{-\eta}, g_0(\eta) = \lambda (1 - \ell^{-\eta}), \theta_0(\eta) = \ell^{-\eta}, \phi_0(\eta) = \ell^{-\eta} \quad (26)$$

First-Order solutions:

From

$$L[u_1] = \eta R_0 \quad (27)$$

We obtain:

$$\left\{ \begin{aligned} f_1(\eta) &= A_1 \ell^{-\eta} + A_2 \eta \ell^{-\eta} \\ g_1(\eta) &= B_1 \ell^{-\eta} + B_2 \eta \ell^{-\eta} \\ \theta_1(\eta) &= C_1 \ell^{-\eta} + C_2 \eta \ell^{-\eta} \\ \phi_1(\eta) &= D_1 \ell^{-\eta} + D_2 \eta \ell^{-\eta} \end{aligned} \right\} \quad (28)$$

Second-Order solutions:

From

$$L[u_2 - u_1] = \eta R_1 \quad (29)$$

We obtain:

$$\left\{ \begin{aligned} f_2(\eta) &= A_3 \ell^{-\eta} + A_4 \eta \ell^{-\eta} + A_5 \eta^2 \ell^{-\eta}, \\ g_2(\eta) &= B_3 \ell^{-\eta} + B_4 \eta \ell^{-\eta} + B_5 \eta^2 \ell^{-\eta}, \\ \theta_2(\eta) &= C_3 \ell^{-\eta} + C_4 \eta \ell^{-\eta} + C_5 \eta^2 \ell^{-\eta}, \\ \phi_2(\eta) &= D_3 \ell^{-\eta} + D_4 \eta \ell^{-\eta} + D_5 \eta^2 \ell^{-\eta}. \end{aligned} \right. \quad (30)$$

And the final truncated Second-Order solutions are:

$$\left. \begin{aligned} f(\eta) &\approx 1 - \ell^{-\eta} + (A_1 + A_3) \ell^{-\eta} + (A_2 + A_4) \eta \ell^{-\eta} + A_5 \eta^2 \ell^{-\eta} \\ g(\eta) &\approx \lambda (1 - \ell^{-\eta}) + (B_1 + B_3) \ell^{-\eta} + (B_2 + B_4) \eta \ell^{-\eta} + B_5 \eta^2 \ell^{-\eta} \\ \theta(\eta) &\approx \ell^{-\eta} + (C_1 + C_3) \ell^{-\eta} + (C_2 + C_4) \eta \ell^{-\eta} + C_5 \eta^2 \ell^{-\eta} \\ \phi(\eta) &\approx \ell^{-\eta} + (D_1 + D_3) \ell^{-\eta} + (D_2 + D_4) \eta \ell^{-\eta} + D_5 \eta^2 \ell^{-\eta} \end{aligned} \right\} \quad (31)$$

where, all constants: A_i, B_i, C_i, D_i , depend on $M, K_p, S, Pr, Rd, Ec, Nb, Nt, Sc$, and γ .

3.2 Quantities of Engineering Interest

In the present study, the skin-friction coefficients in the x and y -directions, together with the local Nusselt number, are formulated based on well-established expressions reported in the existing literature. These engineering parameters are essential for evaluating the wall shear stresses and heat-transfer characteristics of the flow and are defined as follows:

$$C_{fx} = \frac{\tau_{wx}}{\rho u_w^2/2}, C_{fy} = \frac{\tau_{wy}}{\rho u_w^2/2}, Nu_x = \frac{x q_w}{k_f (T_f - T_\infty)} \quad (32)$$

where, τ_{wx} and τ_{wy} represent the wall shear stresses in the x - and y - directions, respectively, while q_w represents the surface heat flux. The mathematical formulations of these physical quantities are based on standard expressions established in the literature and are given as follows:

$$\tau_{wx} = \mu \left. \frac{\partial u}{\partial z} \right|_{z=0}, \tau_{wy} = \mu \left. \frac{\partial v}{\partial z} \right|_{z=0} \quad (33)$$

$$q_w = - \left(k + \frac{16 T_\infty^3 \sigma^*}{3 k^*} \right) \left. \frac{\partial T}{\partial z} \right|_{z=0} \quad (34)$$

4 Results and Discussion

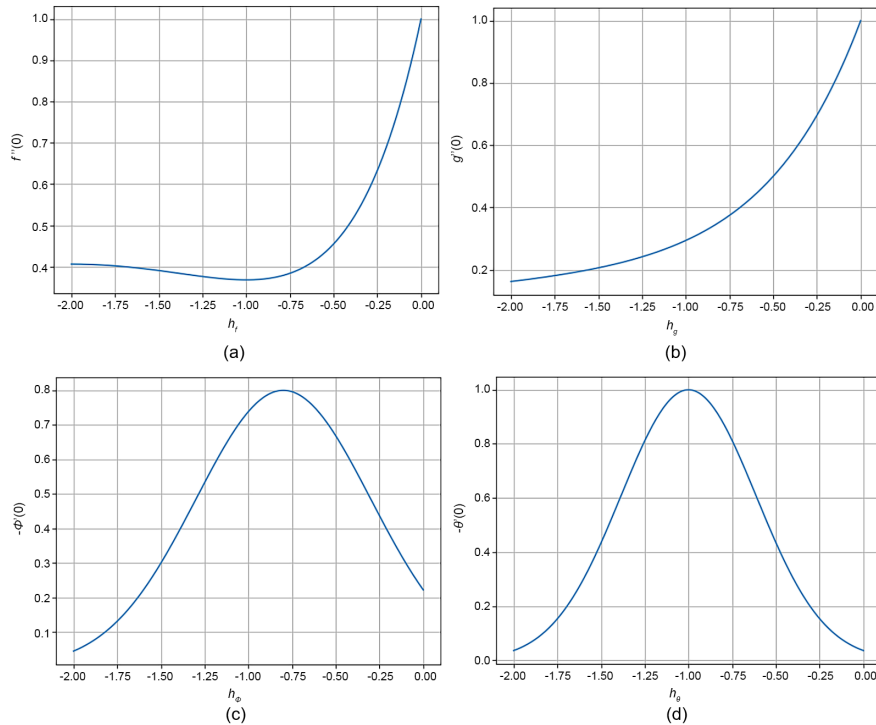


Figure 2. Convergence of the homotopy analysis method (HAM) solution for variations in h_f, h_g, h_θ and h_ϕ

Note: h_f, h_g, h_θ , and h_ϕ are the auxiliary convergence-control parameters of the HAM corresponding to the velocity functions $f(\eta)$ and $g(\eta)$, the temperature function $\theta(\eta)$, and the concentration function $\phi(\eta)$, respectively. Here, $f''(0)$ and $g''(0)$ denote the wall shear parameters, whereas $-\theta'(0)$ and $-\phi'(0)$ represent the local Nusselt and Sherwood number-related quantities, respectively.

In this section, we review the physical properties of the flow, temperature, and concentration fields from the analytic solutions of Eqs. (10) to (13) using the HAM. Since the governing equations are non-linear, a reliable analytical approach is necessary, and HAM is particularly well-suited, because it allows for an effective control of the convergence of the solutions via an appropriate choice of auxiliary parameters h_f , h_g , h_θ and h_ϕ . The convergence of the obtained series solutions is examined through the (h) -curves presented in Figure 2. These curves provide admissible ranges of the convergence-control parameters, ensuring that the series solutions remain bounded and physically meaningful. It is observed that the acceptable convergence regions are approximately: $-1.5 \leq h_f \leq -0.8$, $-1.6 \leq h_g \leq -0.9$, $-1.3 \leq h_\phi \leq -0.7$ and $-1.2 \leq h_\theta \leq -0.6$.

Within these intervals, the solutions exhibit rapid convergence and stability, which is consistent with previous HAM-based studies on nonlinear boundary layer flows [22].

To validate the present results, comparisons are made with the limiting-case results reported by Lone et al. [24]. The velocity profiles in both primary and secondary flow directions exhibit good agreement with previously published results, as seen in Figures 3 and 4, demonstrating the accuracy and reliability of the present analytical solution.

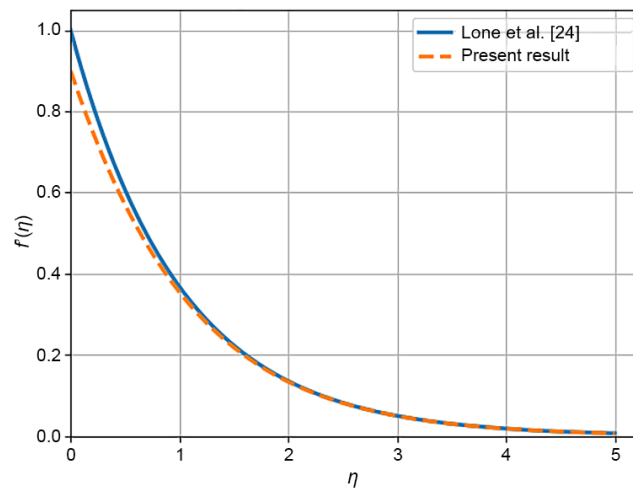


Figure 3. Velocity profile validation (x -direction)

Note: η represents the similarity variable, while $f'(\eta)$ denotes the dimensionless x -direction velocity profile derived from the similarity function $f(\eta)$.

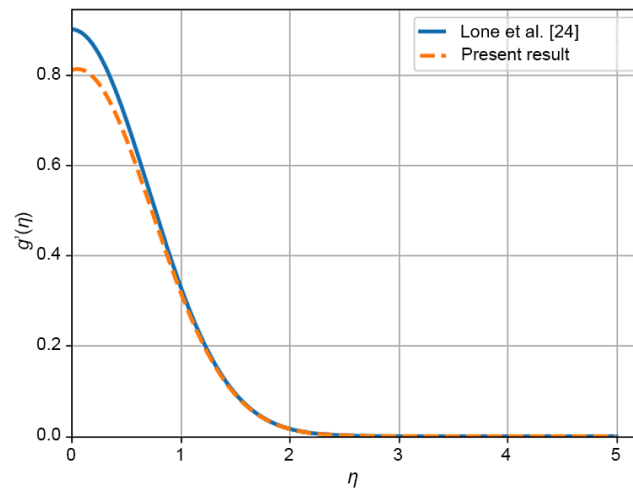


Figure 4. Velocity profile validation (y -direction)

Note: η denotes the similarity variable, and $g'(\eta)$ represents the dimensionless velocity profile in the y -direction, obtained from the derivative of the similarity function $g(\eta)$ with respect to η .

Unless otherwise specified, the parameter values listed in Table 1 are used throughout the numerical analysis. Table 1 summarizes the selected values, typical ranges, physical meanings, and engineering justification of the

governing dimensionless parameters considered in the present study. These parameter values are selected based on commonly adopted ranges reported in the literature for MHD Casson fluid flow and coupled heat and mass transfer studies [24, 25]. The subsequent velocity, temperature, and concentration profiles are therefore evaluated using these parameter settings unless a particular parameter is varied explicitly.

Table 1. Physical justification of governing parameters

Parameter	Typical Range Used	Physical Meaning	Engineering Justification
M	0–2	Magnetic strength	Represents weak to moderately strong magnetic fields in magnetohydrodynamic (MHD) cooling, metallurgy and liquid-metal transport
β	0.5–5	Casson parameter	Covers biological fluids, blood, polymer suspensions and printing inks exhibiting yield stress
K	0–1	Dimensionless porous-resistance parameter	Typical porous substrates used in filtration, catalytic beds and porous heat exchangers
λ	0–1	Thermal relaxation	Represents finite-speed heat propagation in non-Fourier heat conduction
Nb	0.1–0.5	Brownian motion	Corresponds to dilute nanofluids where nanoparticle random motion is important
Nt	0.1–0.5	Thermophoresis	Represents nanoparticle migration caused by temperature gradients
Rd	0–2	Radiation parameter	Typical for moderate radiative heat-transfer applications such as solar collectors and furnaces
Ec	0–0.8	Eckert number	Covers moderate viscous dissipation in high-speed and polymer processing flows
Pr	20.2	Prandtl number	Represents water-based nanofluids at room temperature
Sc	0.6–2	Schmidt number	Typical gases and dilute liquid species used in mass-transfer studies

These values are selected based on commonly adopted ranges in the literature for MHD Casson fluid flow and heat and mass transfer studies [24, 25].

4.1 Results of Velocity Profile

Figures 5–7 demonstrate the effects of the principal governing parameters on the primary and secondary velocity profiles. The velocity characteristics are substantially affected by magnetic forces, fluid rheology, unsteadiness, and resistance offered by the porous medium.

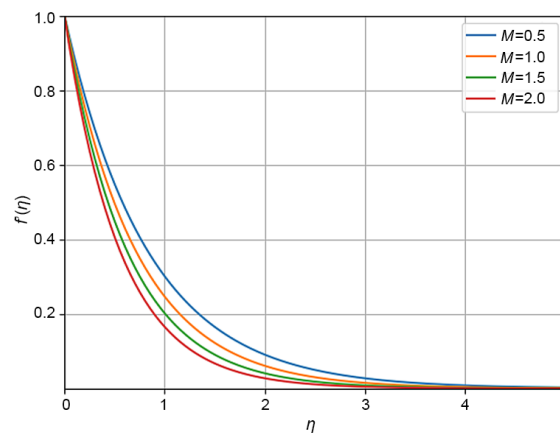


Figure 5. Effect of magnetic field parameter on velocity profile

Note: M is the magnetic field parameter, η represents the similarity variable, while $f'(\eta)$ denotes the dimensionless x -direction velocity profile obtained from the derivative of the similarity function $f(\eta)$ with respect to η .

Figure 5 shows that increasing the magnetic-field strength leads to a marked decrease in both velocity components. This behaviour is attributed to the resistive Lorentz force generated by the interaction between the applied magnetic field and the electrically conducting fluid. The resulting magnetic drag acts against the fluid motion, thereby reducing momentum and suppressing the velocity profiles within the boundary layer.

Figure 6 illustrates the effect of the unsteadiness parameter, for which an increase produces a reduction in both the primary and secondary velocity profiles. This response can be linked to the modification of the momentum balance caused by the time-dependent stretching of the surface, which weakens the fluid motion within the boundary layer.

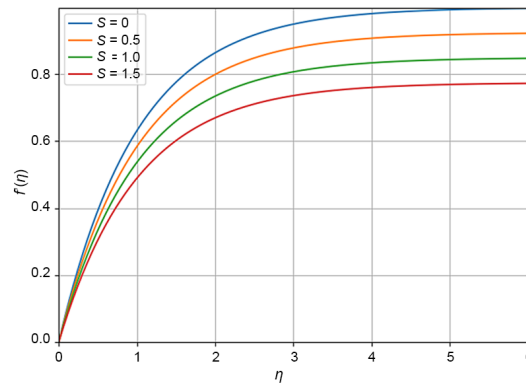


Figure 6. Effect of unsteadiness parameter on the velocity profile

Note: S is the unsteadiness parameter, η represents the similarity variable, while $f'(\eta)$ denotes the dimensionless velocity profile in the x -direction derived from the similarity function $f(\eta)$.

Figure 7 depicts the influence of the dimensionless porous-resistance parameter on the velocity profiles. As the porous resistance increases, both velocity components decrease progressively across the fluid domain. The reduction is caused by the enhanced drag imposed by the porous structure, which opposes fluid movement and consequently restricts momentum transport. This behaviour is directly related to the permeability of the porous medium: higher porous resistance corresponds to lower permeability and, hence, greater opposition to fluid motion, whereas increased permeability allows the fluid to move more readily with reduced resistance. Thus, the decreasing velocity profiles observed in Figure 7 confirm the role of porous-medium resistance as an effective momentum-damping mechanism in the present flow configuration.

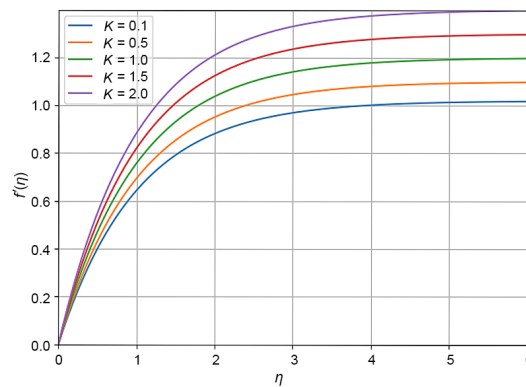


Figure 7. Effect of the dimensionless porous-resistance parameter K on the velocity profile

Note: K denotes the dimensionless porous-resistance parameter, η represents the similarity variable, and $f'(\eta)$ denotes the dimensionless x -direction velocity profile obtained from the derivative of the similarity function $f(\eta)$ with respect to η .

4.2 Results of Temperature

Figures 8–13 present the effects of the principal governing parameters on the temperature distribution. The thermal field is determined by the combined contributions of heat conduction, viscous dissipation, thermal radiation, Brownian motion, thermophoretic effects, and the non-Fourier heat-flux mechanism. Figure 8 indicates that increasing the thermal relaxation parameter reduces the temperature profile. This reduction is associated with the Cattaneo–Christov heat-flux formulation, which incorporates the finite propagation speed of thermal disturbances

and consequently limits the rate of thermal diffusion and heat accumulation within the boundary layer. Figure 9 shows that an increase in the thermophoresis parameter leads to an enhancement of the temperature distribution. The thermophoretic migration of nanoparticles from hotter to colder regions modifies the distribution of thermal energy and contributes to an increase in the thickness of the thermal boundary layer. As depicted in Figure 10, increasing the Brownian motion parameter also raises the temperature profile. The stronger random movement of nanoparticles facilitates microscopic energy transport and promotes greater thermal interaction within the fluid, thereby increasing heat retention in the boundary layer. Figure 11 demonstrates that the Eckert number has a significant positive influence on the temperature distribution. This effect is primarily due to viscous dissipation, whereby mechanical energy associated with fluid motion is transformed into thermal energy. The additional heat generated through this mechanism increases the fluid temperature, particularly in regions characterized by strong velocity gradients. Figure 12 illustrates that an increase in the radiation parameter enhances the temperature profile. Thermal radiation provides an additional mode of energy transfer, increasing the thermal energy available within the fluid and consequently thickening the thermal boundary layer. Finally, Figure 13 shows that increasing the Prandtl number decreases the temperature distribution. Since a larger Prandtl number corresponds to relatively lower thermal diffusivity compared with momentum diffusivity, thermal energy penetrates less effectively into the fluid. This results in a thinner thermal boundary layer and a corresponding reduction in the temperature profile.

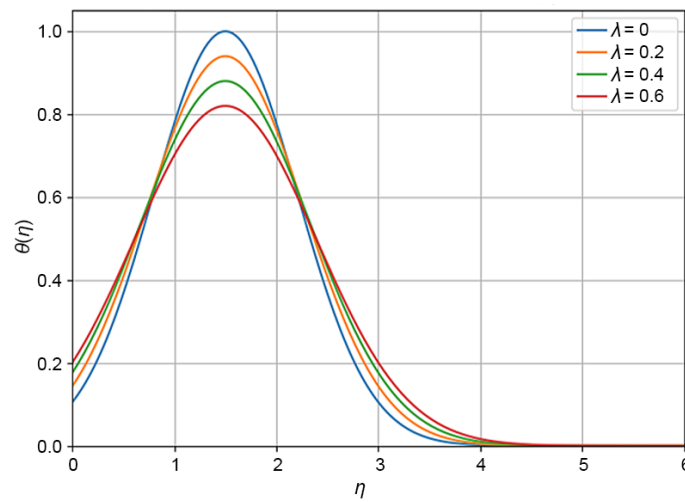


Figure 8. Effect of thermal relaxation parameter on temperature profile

Note: λ denotes the thermal relaxation parameter, η represents the similarity variable, and $\theta(\eta)$ denotes the dimensionless temperature profile obtained from the temperature similarity function.

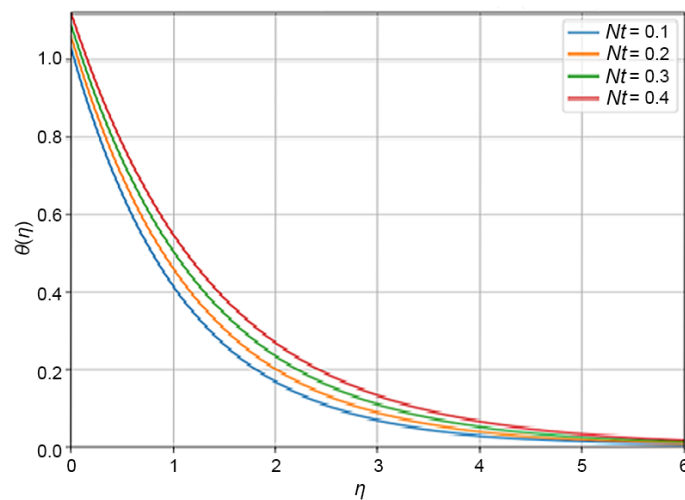


Figure 9. Effect of thermophoresis parameter on temperature distribution

Note: Nt denotes the thermophoresis parameter, η represents the similarity variable, and $\theta(\eta)$ denotes the dimensionless temperature profile obtained from the temperature similarity function.

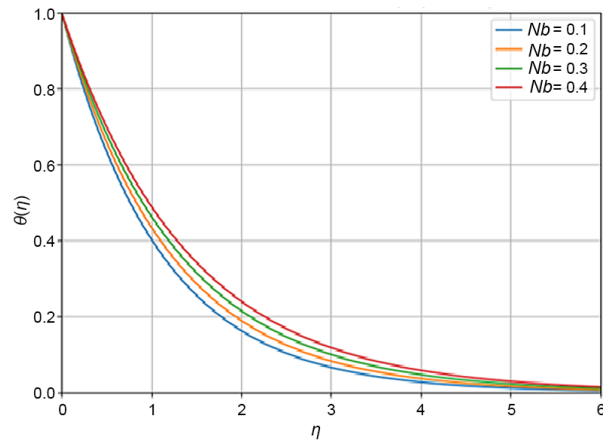


Figure 10. Effect of Brownian motion parameter on temperature distribution

Note: Nb denotes the Brownian motion parameter, η represents the similarity variable, and $\theta(\eta)$ denotes the dimensionless temperature profile obtained from the temperature similarity function.

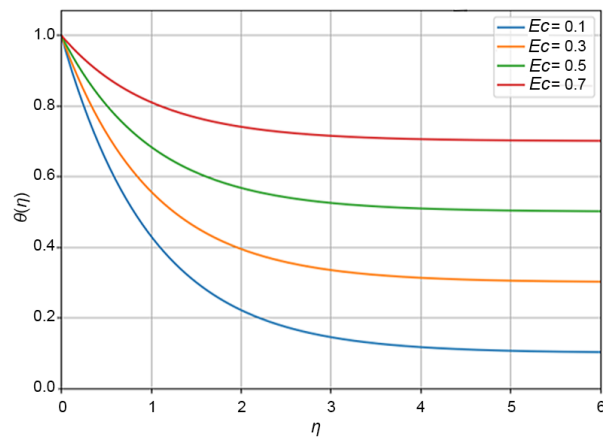


Figure 11. Effect of Eckert number on temperature profile

Note: Ec denotes the Eckert number, η represents the similarity variable, and $\theta(\eta)$ denotes the dimensionless temperature profile obtained from the temperature similarity function.

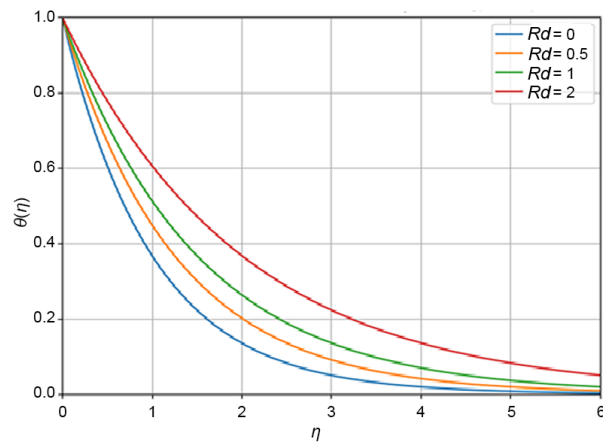


Figure 12. Effect of thermal radiation parameter on temperature profile

Note: Rd denotes the thermal radiation parameter, η represents the similarity variable, and $\theta(\eta)$ denotes the dimensionless temperature profile obtained from the temperature similarity function.

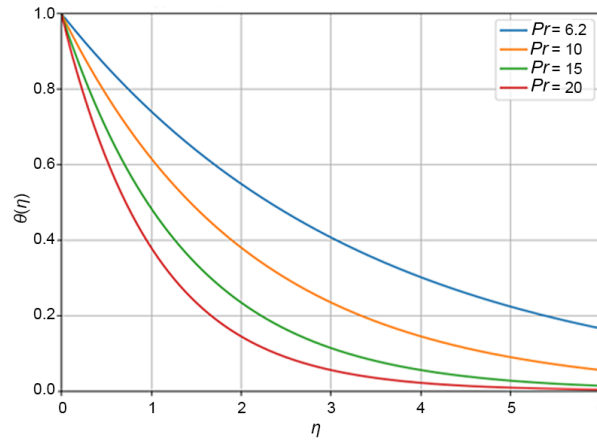


Figure 13. Effect of Prandtl number on temperature profile

Note: Pr denotes the Prandtl number, η represents the similarity variable, and $\theta(\eta)$ denotes the dimensionless temperature profile obtained from the temperature similarity function.

4.3 Results of Concentration

Figures 14 and 15 depict the effects of selected governing parameters on the concentration distribution. The concentration field is primarily influenced by mass diffusivity, thermophoresis, and Brownian motion. As illustrated in Figure 14, an increase in the Schmidt number leads to a reduction in the concentration profile. This trend results from the inverse relationship between the Schmidt number and mass diffusivity, such that a higher Schmidt number corresponds to weaker molecular diffusion. Consequently, the concentration boundary layer becomes thinner, causing a lower concentration distribution within the fluid. Figure 15, however, shows that increasing the thermophoresis parameter enhances the concentration profile. This behaviour is associated with thermophoretic migration, whereby nanoparticles or chemical species move from regions of relatively higher temperature toward regions of lower temperature. The resulting redistribution and accumulation of nanoparticles within the concentration boundary layer consequently increase the concentration distribution.

Table 2 illustrates the effect of the thermophoresis parameter on the skin-friction coefficients, Nusselt number, and Sherwood number. The corrected results indicate that the skin-friction coefficients along the x - and y -directions remain unchanged with increasing thermophoresis parameter. This result agrees with the mathematical structure of the governing model, since the thermophoresis parameter is absent from both transformed momentum equations. Consequently, variations in the thermophoresis parameter do not alter the velocity profiles and, hence, have no influence on the corresponding wall shear stresses or skin-friction coefficients. Likewise, the local Nusselt number remains essentially constant for the range of thermophoresis parameter values considered. This behaviour is also consistent with the transformed energy equation, in which the thermophoresis parameter does not explicitly occur. Thus, changes in thermophoresis do not affect the temperature distribution or its wall-normal gradient within the present formulation. In contrast, the Sherwood number exhibits a progressive decline as the thermophoresis parameter increases, demonstrating its direct influence on nanoparticle mass transfer. This trend arises from the explicit inclusion of the thermophoretic parameter in the transformed concentration equation through the thermophoretic transport term. As the thermophoresis parameter increases, the nanoparticle concentration profile is modified, leading to a reduction in the concentration gradient at the wall and, consequently, a lower nanoparticle mass-transfer rate as measured by the Sherwood number.

Table 2. Effect of thermophoresis parameter on skin friction, Nusselt number, and Sherwood number

Nt	C_{fx}	C_{fy}	Nu_x	Sh_x
0.1	1.018	0.744	1.942	1.612
0.2	1.018	0.744	1.942	1.462
0.3	1.018	0.744	1.942	1.331
0.4	1.018	0.744	1.942	1.215

Note: Nt is the thermophoresis parameter, C_{fx} and C_{fy} are the local skin friction coefficients along the x - and y -directions, respectively, while Nu_x and Sh_x represent the local Nusselt and Sherwood numbers, respectively.

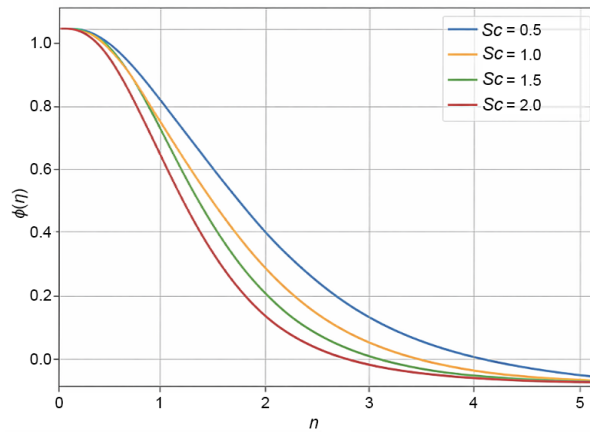


Figure 14. Effect of Schmidt number on concentration profile

Note: Sc denotes the Schmidt number, η represents the similarity variable, and $\phi(\eta)$ denotes the dimensionless concentration profile obtained from the concentration similarity function.

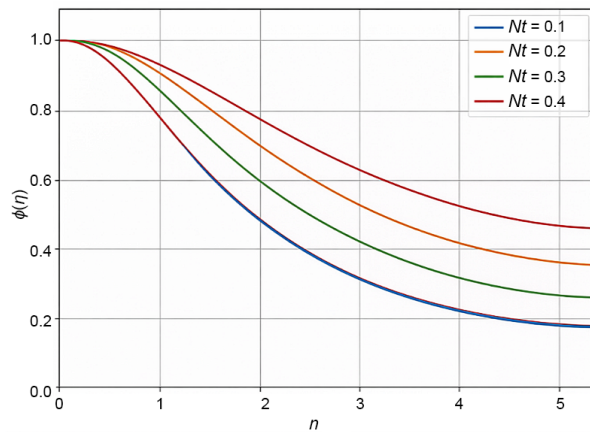


Figure 15. Effect of thermophoresis parameter on concentration profile

Note: Nt denotes the thermophoresis parameter, η represents the similarity variable, and $\phi(\eta)$ denotes the dimensionless concentration profile obtained from the concentration similarity function.

4.4 Engineering Significance of the Present Study

The present study offers significant insights into the transport characteristics of electrically conducting Casson nanofluids subjected to the combined effects of magnetic fields, porous media, thermal radiation, viscous dissipation, Brownian motion, thermophoresis, and thermal relaxation. The observed decrease in fluid velocity with increasing magnetic-field strength indicates that magnetic forces can effectively regulate the motion of electrically conducting fluids, with potential applications in electromagnetic and continuous casting, metallurgical processing, and other manufacturing operations requiring precise flow control. Similarly, the effect of the porous-medium parameter provides useful information for controlling fluid transport through permeable structures, which is relevant to the design and optimisation of filtration units, catalytic reactors, and porous heat-exchange systems. The rise in fluid temperature associated with thermal radiation and viscous dissipation has important implications for thermal management in polymer extrusion, glass-fibre manufacturing, coating processes, and other industrial operations involving non-Newtonian fluids. In contrast, the reduction in temperature observed with increasing thermal relaxation parameter highlights the capability of the Cattaneo–Christov heat-flux model to describe finite-speed heat propagation more realistically than the classical Fourier model, particularly in systems involving rapid thermal transients, microscale heat transfer, and high-temperature processing. Brownian motion and thermophoretic diffusion also play important roles in nanoparticle migration, thereby influencing the overall heat- and mass-transfer characteristics of the fluid. Such behaviour is relevant to biomedical transport, targeted drug-delivery systems, therapeutic cooling, and other applications where precise control of nanoparticle distribution is essential. Enhanced nanoparticle transport may further improve thermal performance in electronic cooling, compact heat exchangers, and advanced refrigeration technologies. From an energy-technology perspective, the proposed analytical framework contributes to a better

understanding of coupled heat and mass transport in non-Newtonian nanofluids, with potential applications in solar thermal collectors, geothermal energy systems, nuclear reactor cooling, and other advanced energy technologies. Overall, the findings provide useful theoretical and practical guidance for the design, optimisation, and control of engineering systems involving electrically conducting non-Newtonian nanofluids exposed to complex magnetic and thermal conditions.

To further assess the reliability of the present HAM solution, the computed dimensionless velocity distributions $f'(\eta)$ and $g'(\eta)$ were compared with the graphical results reported by Lone et al. [24] under identical limiting conditions. The approximate quantitative comparison presented in Tables 3 and 4 demonstrates excellent agreement between the two solutions. The largest deviations occur near the wall ($\eta = 0$), while the differences decrease rapidly with increasing similarity variable, becoming practically negligible in the outer boundary layer. These small discrepancies are attributed to graphical data extraction, differences in the truncation order of the HAM series, and numerical precision. Overall, the close agreement confirms the correctness, convergence, and reliability of the present analytical formulation.

Table 3. Approximate graphical validation of the present homotopy analysis method (HAM) solution with Lone et al. [24] for $f'(\eta)$ (Figure 3)

η	Lone et al. [24]	Present HAM	Absolute Difference	Relative Error (%)
0.0	1.000	0.900	0.100	10.00
0.5	0.590	0.565	0.025	4.24
1.0	0.380	0.365	0.015	3.95
1.5	0.230	0.223	0.007	3.04
2.0	0.135	0.132	0.003	2.22
3.0	0.048	0.047	0.001	2.08
4.0	0.018	0.018	<0.001	<1.0
5.0	0.008	0.008	<0.001	<1.0

Note: η represents the similarity variable, while $f'(\eta)$ denotes the dimensionless x -direction velocity profile obtained from the derivative of the similarity function $f(\eta)$ with respect to η .

Table 4. Approximate graphical validation of the present homotopy analysis method (HAM) solution with Lone et al. [24] for $g'(\eta)$ (Figure 4)

η	Lone et al. [24]	Present HAM	Absolute Difference	Relative Error (%)
0.0	0.910	0.810	0.100	10.99
0.5	0.760	0.720	0.040	5.26
1.0	0.400	0.390	0.010	2.50
1.5	0.120	0.115	0.005	4.17
2.0	0.020	0.019	0.001	5.00
3.0	0.001	0.001	<0.001	<1.0

Note: η denotes the similarity variable, and $g'(\eta)$ represents the dimensionless velocity profile in the y -direction.

5 Conclusion

The present study analytically examines the unsteady three-dimensional MHD flow of a two-phase Casson nanofluid over a bidirectionally stretching surface, incorporating the effects of porous media, thermal radiation, viscous dissipation, Brownian motion, thermophoresis, double-diffusive transport, and the Cattaneo–Christov heat flux model. The two-phase formulation accounts for nanoparticle migration, while the HAM is employed to derive convergent analytical solutions. The validity and accuracy of the proposed model are established through graphical and quantitative comparisons with previously published results. The findings indicate that stronger magnetic-field intensity and porous-medium resistance significantly reduce the velocity profiles, whereas increased thermal radiation and viscous dissipation enhance the temperature distribution.

Brownian motion and thermophoresis have significant effects on nanoparticle concentration and distribution within the flow. In comparison with the classical Fourier heat conduction model, the Cattaneo–Christov formulation predicts lower temperature profiles because it accounts for thermal relaxation and the finite speed of heat propagation. The ranges of the governing parameters considered in this study are representative of realistic engineering conditions, thereby supporting the practical relevance and applicability of the proposed model.

Overall, the developed analytical model provides a reliable framework for predicting the coupled momentum, heat, and mass transport characteristics of electrically conducting Casson nanofluids. The results offer useful insights

into several engineering and technological applications, including polymer processing, cooling systems, metallurgical manufacturing, biomedical transport, and advanced thermal energy systems. Future investigations could extend the present formulation to hybrid nanofluids, variable thermophysical properties, and chemically reactive flow systems.

The engineering performance indicators further reveal that the thermophoresis parameter primarily influences nanoparticle mass transfer. Specifically, increasing the thermophoresis parameter leads to a progressive reduction in the Sherwood number, whereas the skin-friction coefficients in both directions and the local Nusselt number remain essentially unchanged within the range of parameters considered. This indicates that, in the present formulation, thermophoresis predominantly affects the concentration field and associated mass-transfer rate without producing a significant change in the wall shear stress or heat-transfer rate. Finally, the study enhances the understanding of the complex interactions among fluid motion, heat transfer, and mass transport in non-Newtonian MHD flows. These findings provide a useful basis for the analysis, design, and optimisation of advanced industrial and engineering processes involving electrically conducting nanofluids.

Limitations of the Study

Despite the significant contributions of the present study, several limitations should be acknowledged. The analysis is restricted to laminar three-dimensional boundary-layer flow, and therefore does not account for possible turbulence effects. The induced magnetic field is neglected under the assumption of a low magnetic Reynolds number, which may not be applicable to all practical flow conditions. In addition, the Casson fluid model is used exclusively to characterize the non-Newtonian behaviour, whereas other constitutive models may produce different hydrodynamic and heat-transfer characteristics. The analytical solution obtained through the HAM is also truncated at a finite order, which may introduce minor approximation errors despite the use of suitable convergence-control parameters. Furthermore, the thermophysical properties are assumed to be constant, although they may vary with temperature and nanoparticle concentration in real applications. The formulation does not consider chemical reactions or external body forces such as gravity and buoyancy-driven convection. Finally, the nanoparticle suspension is assumed to be dilute and uniformly dispersed, with particle agglomeration, sedimentation, and inter-particle interactions neglected. These assumptions should therefore be considered when interpreting the results and assessing the applicability of the model to more complex practical systems.

Author Contributions

Conceptualization, S.I. and A.M.; methodology, S.I. and A.M.; software, I.A.; validation, S.I., I.A., and A.M.; formal analysis, A.M. and I.A.; investigation, S.I. and I.A.; resources, S.I.; data curation, I.A.; writing original draft preparation, S.I., I.A., and A.M.; writing review and editing, S.I., I.A., and A.M.; visualization, I.A.; supervision, S.I.; project administration, S.I. All authors have read and agreed to the published version of the manuscript.

Data Availability

Not applicable.

Conflict of Interest

The authors declare no conflicts of interest

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